



SCREEN TO SKY

Development Debrief & Architecture Report

ST-500 Series — Flight School Backbone Build

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Table of Contents

1. **Executive Summary** — What We Built
2. **Architecture Overview** — Stack, Services, Data Flow
3. **Database Schema** — All Models & Relationships
4. **Telemetry Pipeline** — Upload → Parse → Analyze → Score → Debrief
5. **17-Lesson Path View** — Curriculum Structure & Gate Logic
6. **Scoring Engine** — C172 Constants, 1-5 Scale, Ramped Standards
7. **Authentication & Authorization** — Roles, Admin Bypass
8. **API Reference** — All Endpoints
9. **UI Pages & Components** — Frontend Architecture
10. **Telemetry Recorder (FlyWithLua v4)** — X-Plane Integration
11. **Lesson Readiness Map** — What's Ready vs TODO
12. **What Remains** — Next-Agent Task List
13. **Appendix A:** Telemetry Field Reference
14. **Appendix B:** Admin Operating Instructions

1. Executive Summary

Screen to Sky (S2S) is a PPL flight training platform that bridges X-Plane flight simulation with structured curriculum delivery, automated flight grading, and progressive lesson unlocking. This document covers the **backbone build** (ST-500 series) — the infrastructure that content specialists will populate.

17

Lessons Built

4

Check Flight Gates

63

Telemetry Channels

14

API Endpoints

What Was Accomplished

- **Full-stack web application** with authentication, role-based access (STUDENT/ADMIN)
- **17-lesson career path** organized across 5 arcs with gated check flights at L4, L7, L10, L14
- **Telemetry pipeline:** CSV upload → flexible parsing → flight segmentation → 1-5 scoring → debrief visualization
- **FlyWithLua v4 recorder:** 63-channel telemetry capture at 10Hz, validated against Tacview (99.97% altitude correlation)
- **Ground school quiz scaffold** with gate logic (quiz + flight score + debrief = unlock)
- **Admin tools:** force-complete, readiness flags, raw telemetry debug panels
- **Content readiness system:** 5 flags + 4 production notes per lesson for specialist handoff

Key Principle: This build is the *flight school backbone*, not the final content universe. Every lesson, quiz, scoring rubric, and media asset has clean interfaces for future agents/specialists to populate.

2. Architecture Overview

Technology Stack

Layer	Technology	Purpose
Framework	Next.js 14 (App Router)	Server/client component architecture
Styling	Tailwind CSS	Utility-first responsive design
Animation	Framer Motion	Path View transitions, tab animations
Database	PostgreSQL via Prisma ORM	Structured data persistence
Auth	NextAuth.js v4 (JWT)	Credentials provider, role-based sessions
Storage	AWS S3	Telemetry CSV file storage via presigned URLs
Charts	Recharts	Flight path, altitude, speed, controls visualization

Data Flow: Flight → Score → Debrief

```
Student flies in X-Plane (FlyWithLua v4 records CSV at 10Hz)
↓
Student uploads CSV to S2S web app (manual upload)
↓
Telemetry Parser (lib/telemetry-parser.ts)
  → Flexible schema: handles 38+ expected columns, ignores missing
  → Computes summary stats (max alt, max IAS, landings, T&Gs)
↓
Flight Analyzer (lib/flight-analyzer.ts)
  → Segments flight into phases: Preflight, Taxi, Takeoff, Climb,
    Downwind, Approach, Landing, Touch-and-Go, Full Stop
  → Flags uncertain segments as "NEEDS_REVIEW"
↓
Scoring Engine (lib/scoring-engine.ts)
  → Evaluates against C172 constants (Vr=55, Vy=74, App=65)
  → Outputs 1-5 score with breakdown + coaching notes
↓
Database: Session + Attempts + Score stored
↓
Debrief Page: Charts + Score Card + Coaching Notes
  → Student marks debrief as reviewed → gate check
```

Service Modules

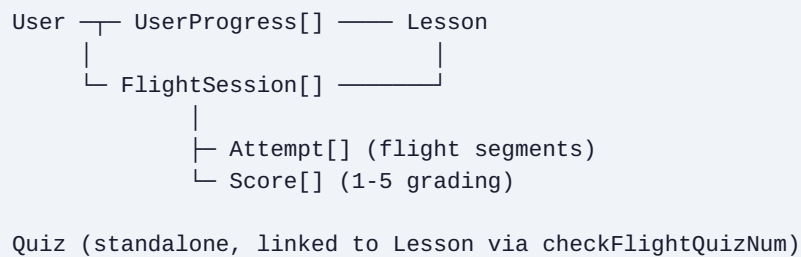
Service	File	Purpose
---------	------	---------

Telemetry Parser	<code>lib/telemetry-parser.ts</code>	CSV → structured rows + summary stats
Flight Analyzer	<code>lib/flight-analyzer.ts</code>	Rows → flight segments (taxi, takeoff, etc.)
Scoring Engine	<code>lib/scoring-engine.ts</code>	Segments + rows → 1-5 score + coaching notes
Auth Options	<code>lib/auth-options.ts</code>	NextAuth config with ADMIN/STUDENT roles
S3 Utilities	<code>lib/s3.ts</code>	Presigned upload/download for telemetry CSVs
Prisma Client	<code>lib/prisma.ts</code>	Database connection singleton

3. Database Schema

7 core models + 3 NextAuth support models. All relationships use cascading deletes.

Entity Relationship Diagram



Model: User

Field	Type	Notes
id	String (cuid)	Primary key
email	String	Unique
passwordHash	String	bcrypt hashed
role	String	"ADMIN" or "STUDENT" (default)
name	String?	Optional display name

Model: Lesson

Field	Type	Notes
lessonNum	Int	1-17, unique
arcNum	Int	1-5 (arc grouping)
title	String	e.g., "Long Walk"
description	Text	Lesson overview
objectives	JSON	String array of learning objectives
briefContent	Text?	Instructor briefing content
isCheckFlight	Boolean	True for L4, L7, L10, L14
checkFlightQuizNum	Int?	250-253 for check flights

Content Readiness Flags		
flightSchoolReady	Boolean	Flight scenario defined
groundSchoolReady	Boolean	Ground school content available
quizReady	Boolean	Quiz questions populated
scoringReady	Boolean	Scoring rubric calibrated
mediaReady	Boolean	Videos/media assets produced
Production Notes (for Content Agents)		
groundSchoolProductionNotes	Text?	What ground school content is needed
flightRubricProductionNotes	Text?	What scoring logic is needed
mediaProductionNotes	Text?	What media assets are needed
scoringTuningNotes	Text?	Scoring calibration notes

Model: FlightSession

Field	Type	Notes
userId / lessonId	FK	Links to User and Lesson
csvFilename	String	Original upload filename
cloudStoragePath	String?	S3 key for stored CSV
parsedData	JSON	Sampled telemetry rows + summary
rowCount	Int	Total data points in CSV
durationSec	Float	Flight duration in seconds

Model: Score

Field	Type	Notes
overallScore	Int	1-5 ONLY (no percentages)
breakdown	JSON	Per-metric scores: {rotationSpeed, climbSpeed, patternAlt, approachSpeed, coordination}

Model: UserProgress

Field	Type	Notes
userId + lessonId	Composite Unique	One progress record per user per lesson
flightScore	Int?	Best flight score (1-5)

quizPassed	Boolean	True if quiz score >= required
debriefViewed	Boolean	True if student viewed debrief
status	String	LOCKED AVAILABLE IN_PROGRESS NEEDS_DEBRIEF PASSED

Model: Quiz

Field	Type	Notes
quizNum	Int	250 (Arc 1), 251 (Arc 2), 252 (Arc 3), 253 (Arc 4)
questions	JSON	Array of {question, options[], correctIndex, explanation}
requiredScore	Int	Minimum % to pass (default 80)

4. Telemetry Pipeline

4.1 Upload Flow

1. Student navigates to lesson → Flight tab
2. Selects CSV file from desktop (generated by FlyWithLua v4)
3. Client reads CSV content, requests presigned S3 URL
4. CSV uploaded to S3 for permanent storage
5. CSV content sent to `/api/sessions` for processing

4.2 Flexible Parser

Design Principle: The parser NEVER crashes on missing columns. Unknown columns are captured as extras. This future-proofs the system for telemetry recorder updates.

38 Expected Columns: `t_sec`, `sim_time`, `lat`, `lon`, `alt_msl`, `alt_agl`, `ias_kts`, `gs_kts`, `vs_fpm`, `pitch_deg`, `roll_deg`, `hdg_deg`, `aileron`, `elevator`, `rudder`, `throttle`, `mixture`, `flap_ratio`, `elv_trim`, `rpm`, `mp_inhg`, `on_ground`, `parking_brake`, `g_normal`, `slip_deg`, `stall_warn`, `chock_left`, `chock_right`, `tiedown_left`, `tiedown_right`, `tiedown_tail`, `pitot_cover`, `head_yaw`, `head_pitch`, `view_type`, `beacon_on`, `battery_on`, `avionics_on`

4.3 Flight Segmentation

Segment Type	Detection Logic	Confidence
PREFLIGHT	<code>on_ground=1</code> , <code>gs < 5 kts</code>	HIGH
TAXI	<code>on_ground=1</code> , <code>gs >= 5 kts</code> , IAS not accelerating	HIGH
TAKEOFF_ROLL	<code>on_ground=1</code> , IAS increasing > 20 kts delta	HIGH
LIFTOFF	<code>on_ground</code> transitions $1 \rightarrow 0$	HIGH
CLIMB	<code>vs_fpm > 300</code> , <code>alt_agl < 800</code>	HIGH
DOWNWIND	<code>alt_agl</code> 800-1200 ft (pattern altitude band)	MEDIUM
APPROACH	<code>vs_fpm < -300</code> , <code>alt_agl < 800</code>	HIGH
LANDING	<code>on_ground</code> transitions $0 \rightarrow 1$, followed by <code>gs</code> $\rightarrow 0$	HIGH
TOUCH_AND_GO	Landing + airborne again within 15 sec	HIGH
NEEDS_REVIEW	Segment doesn't match known patterns	LOW

Design Rule: If segmentation is uncertain, it's flagged as NEEDS_REVIEW rather than guessing. The system never invents confidence it doesn't have.

4.4 Summary Statistics

Computed automatically from every upload:

- Max altitude (MSL & AGL), max IAS, max groundspeed
- Max/min vertical speed, max RPM, max/min G-load
- Total time on ground vs in air
- Landing count, touch-and-go count

5. 17-Lesson Path View

5.1 Curriculum Structure

Arc	Name	Lessons	Theme
1	Mechanical Intimacy	L1-L4	Aircraft control & pattern foundation at KHMP
2	Pattern Master	L5-L7	Advanced pattern work, wind correction
3	Departure Protocols	L8-L10	Navigation, radio work, controlled airspace
4	Emergency & Elements	L11-L14	Engine failures, weather, night flying
5	Cross-Country Solo	L15-L17	Flight planning, dual XC, solo XC

5.2 Lesson Details

L#	Title	Type	Key Objectives
1	Long Walk	Flight	High-speed taxi, first circuit, Vr/Vy basics
2	Configurator	Flight	Flaps/trim management, configuration changes in pattern
3	Edge of Envelope	Flight	Slow flight, stalls, steep turns, walk-around
4	Arc 1 Check	CHECK	5 T&Gs + emergency landing Rwy 06, Quiz 250
5	The Box	Flight	Rectangular pattern, ground reference
6	Wind Whisperer	Flight	Crosswind landings, wind correction angles
7	Arc 2 Check	CHECK	Pattern mastery evaluation, Quiz 251
8	Breaking Free	Flight	First departure from KHMP, navigation basics
9	Compass Rose	Flight	VOR navigation, pilotage, dead reckoning
10	Arc 3 Check	CHECK	Navigation proficiency, Quiz 252
11	Dead Stick	Flight	Engine failure procedures, best glide (68 kts)
12	Weather Wise	Flight	Weather systems, go/no-go decision making
13	Night Owl	Flight	Night flying procedures
14	Arc 4 Check	CHECK	Emergency mastery, Quiz 253
15	Flight Plan	Flight	Cross-country planning, weight & balance
16	Dual XC	Flight	Cross-country with instructor reference

5.3 Gate Logic

Student Unlock Condition:

Previous Lesson: `flight_score >= 3 AND quiz_passed == true AND debrief_viewed == true`

Admin Bypass: Admins see all lessons as AVAILABLE regardless of progress. Can force-complete any lesson.

5.4 Node States

State	Visual	Meaning
LOCKED	Grayed out, not clickable	Prerequisites not met
AVAILABLE	Highlighted, clickable	Ready to start
IN_PROGRESS	Active indicator	Flight uploaded, not yet passed
NEEDS_DEBRIEF	Debrief icon	Flight scored ≥ 3 , debrief not viewed
PASSED	Checkmark, green	All gates cleared
PLACEHOLDER	Dotted outline	Content not ready (<code>flightSchoolReady = false</code>)

6. Scoring Engine

6.1 Scale Definition

Score	Label	Meaning
1	Rough	Dangerous deviation, needs immediate correction
2	Needs Work	Significant deviation, not safe for progression
3	Solid	Safe and acceptable — minimum passing score
4	Good	Consistent, minor deviations only
5	Nailed It	Textbook execution

6.2 C172 Reference Constants

Parameter	Value	Tolerance (Score 3)
Rotation Speed (Vr)	55 kts	±5 kts
Best Rate of Climb (Vy)	74 kts	±10 kts
Downwind Speed	90 kts	±10 kts
Approach Speed	65 kts	±8 kts
Best Glide	68 kts	±5 kts

6.3 KHMP Airport Constants

Parameter	Value
Field Elevation	882 ft MSL
Pattern Altitude	1,882 ft MSL (1,000 ft AGL)
Standard Runway	24 (all normal operations)
Emergency Runway	06 (emergency/short-field only)

6.4 Scoring Metrics (Currently Implemented)

Metric	Method	Status
Rotation Speed	IAS at on_ground 1 → 0 transition vs Vr (55)	LIVE

Climb Speed	Average IAS during CLIMB segments vs Vy (74)	LIVE
Pattern Altitude	Average AGL during DOWNWIND vs 1,000 ft	LIVE
Approach Speed	Average IAS during APPROACH vs 65 kts	LIVE
Coordination	Average slip_deg during airborne flight	LIVE
Flap Timing	Detect flap deployment vs segment position	TODO
Stall Recovery	stall_warn + pitch/airspeed analysis	TODO
Walk-Around	Static item toggle sequence + timing	TODO
T&G Consistency	Cross-lap comparison of speeds/altitudes	TODO
Emergency Landing	Engine-off approach to Rwy 06	TODO

6.5 Deviation Scoring Function

```
scoreDeviation(actual, target, tolerance):
    deviation = |actual - target|
    if deviation <= tolerance × 0.2 → Score 5 (Nailed It)
    if deviation <= tolerance × 0.5 → Score 4 (Good)
    if deviation <= tolerance × 1.0 → Score 3 (Solid)
    if deviation <= tolerance × 2.0 → Score 2 (Needs Work)
    else → Score 1 (Rough)
```

7. Authentication & Authorization

7.1 Roles

Role	Capabilities
STUDENT	View unlocked lessons, upload telemetry, take quizzes, view debriefs, track progress
ADMIN	Everything STUDENT can do PLUS: view all lessons regardless of lock, force-complete, toggle readiness flags, view raw telemetry debug panels, manage students

7.2 Seeded Accounts

Email	Password	Role
admin@s2s.com	S2SA123	ADMIN

7.3 Auth Flow

- JWT strategy (no database sessions for performance)
- Middleware protects /dashboard, /lesson/*, /admin routes
- Non-admin users redirected away from /admin
- Session includes user id and role in JWT token

8. API Reference

Endpoint	Method	Auth	Purpose
/api/signup	POST	None	Register new student (email, password, name)
/api/auth/login	POST	None	Validate credentials, return user info
/api/auth/[...nextauth]	*	—	NextAuth session management
/api/lessons	GET	Required	All 17 lessons with computed status
/api/lessons/[num]	GET	Required	Single lesson + progress + sessions + quiz
/api/sessions	POST	Required	Upload CSV → parse → score → create session
/api/sessions/[id]	GET	Required	Session details with scores & segments
/api/quiz/submit	POST	Required	Submit quiz answers, get results
/api/progress/debrief-viewed	POST	Required	Mark debrief reviewed, check completion
/api/upload/presigned	POST	Required	Get S3 presigned URL for CSV upload
/api/admin/students	GET	ADMIN	List all students with progress
/api/admin/force-complete	POST	ADMIN	Force-complete a lesson for a student
/api/admin/lesson-flags	PATCH	ADMIN	Toggle content readiness flags

9. UI Pages & Components

9.1 Page Map

Route	Component	Description
/	Redirect	→ /dashboard (authenticated) or /login
/login	LoginForm	Login/signup with email/password
/dashboard	DashboardClient	17-lesson Path View with arc grouping
/lesson/[num]	LessonClient	Brief, Quiz, Flight, Debrief tabs
/admin	AdminClient	Student management, readiness flags

9.2 Lesson Page Tabs

Tab	Component	Content
Brief	BriefTab	Description, objectives, hub location, instructor briefing
Quiz	QuizTab	Multiple choice (check flights only), pass/fail gate
Flight	FlightTab	CSV upload, previous sessions list, upload results
Debrief	DebriefTab + DebriefCharts	Score card, coaching notes, flight path/alt/speed charts, admin debug panel

9.3 Debrief Charts

- **Flight Path (2D):** Lat/Lon scatter plot with segment color coding
- **Altitude Profile:** MSL + AGL over time, pattern altitude reference line
- **Speed Profile:** IAS over time, Vr/Vy/Vapproach reference lines
- **Control Inputs:** Aileron, elevator, rudder over time
- **Summary Stats:** Max alt, max IAS, duration, landings, T&Gs

10. Telemetry Recorder (FlyWithLua v4)

10.1 Overview

The `s2s_telemetry_recorder_v4.lua` FlyWithLua script runs inside X-Plane 12 and records 63 channels of flight data at 10Hz to CSV. It was developed iteratively across 4 versions:

Version	Change	Status
v1	Initial 29-column recorder	Quarantined (bad dataref)
v2	Fixed dataref, added <code>safe_dataref</code> wrapper	Working, but engine data = 0
v3	Added REP engine datarefs, head tracking	Working, static items = 0
v4	Correct REP static item datarefs, throttle/mixture fix	CURRENT

10.2 Validation Results

Tacview Cross-Reference (May 8, 2026):
Altitude correlation: **r = 0.9997** (MAE: 2.4 meters)
Airspeed correlation: **r = 0.9979** (MAE: 0.87 knots)
Pitch correlation: **r = 0.985**
Roll correlation: **r = 0.973**

10.3 Key Data Categories (63 Channels)

Category	Channels	Examples
Position	6	lat, lon, alt_msl, alt_agl, vs_fpm, gs_kts
Attitude	4	pitch_deg, roll_deg, hdg_deg, alpha_deg
Speeds	2	ias_kts, gs_kts
Controls	7	aileron, elevator, rudder, throttle, mixture, flap_ratio, elv_trim
Engine (REP)	5	rpm, mp_inhg, oil_temp, oil_psi, fuel_flow
Walk-Around	7	chock_left/right, tiedown_L/R/tail, pitot_cover, towbar
Lights/Switches	6	beacon, nav, strobe, landing, battery, avionics
Head Tracking	4	head_yaw, head_pitch, head_roll, view_type
Flight Safety	5	g_normal, g_side, slip_deg, stall_warn, on_ground

10.4 REP-Specific Datarefs Discovered

Item	Dataref Path	Logic
Left Chock	simcoders/rep/landinggear/tires/show_chocks_0	1=Present, 0=Removed
Right Chock	simcoders/rep/landinggear/tires/show_chocks_2	1=Present, 0=Removed
Left Tiedown	simcoders/rep/tiedown/show_0	1=Present, 0=Removed
Right Tiedown	simcoders/rep/tiedown/show_1	1=Present, 0=Removed
Tail Tiedown	simcoders/rep/tiedown/show_2	1=Present, 0=Removed
Pitot Cover	simcoders/rep/pitot/cover	1=Present, 0=Removed
Towbar	simcoders/rep/landinggear/tires/show_towbar	1=Present, 0=Removed
Engine RPM	simcoders/rep/engine/rpm	0-2700 RPM
Oil Temp	simcoders/rep/engine/oiltemp_degF	32-250°F
Oil Pressure	simcoders/rep/engine/oilpsi	0-100 PSI

11. Lesson Readiness Map

L#	Title	Flight	Ground	Quiz	Scoring	Media
1	Long Walk	✓	✗	✗	✓	✗
2	Configurator	✓	✗	✗	✗	✗
3	Edge of Envelope	✓	✗	✗	✗	✗
4	Arc 1 Check	✓	✗	✓	✗	✗
5	The Box	✗	✗	✗	✗	✗
6	Wind Whisperer	✗	✗	✗	✗	✗
7	Arc 2 Check	✗	✗	✓	✗	✗
8	Breaking Free	✗	✗	✗	✗	✗
9	Compass Rose	✗	✗	✗	✗	✗
10	Arc 3 Check	✗	✗	✓	✗	✗
11	Dead Stick	✗	✗	✗	✗	✗
12	Weather Wise	✗	✗	✗	✗	✗
13	Night Owl	✗	✗	✗	✗	✗
14	Arc 4 Check	✗	✗	✓	✗	✗
15	Flight Plan	✗	✗	✗	✗	✗
16	Dual XC	✗	✗	✗	✗	✗
17	Solo XC	✗	✗	✗	✗	✗

Summary: 4 of 85 readiness flags are ✓ (4.7%). Arc 1 has the most complete structure. All other arcs are placeholder scaffolds awaiting content specialists.

12. What Remains — Next-Agent Task List

12.1 HIGH Priority (Needed for Arc 1 Completion)

Story	Title	Description
ST-520	Ground School: L1-L3	Video content for cockpit tour, flaps/trim, stalls/turns
ST-523	Quiz 250 Expansion	Expand Arc 1 quiz from seed questions to full 10-15 question bank
ST-530	L2 Scoring: Flap Timing	Detect flap deployment timing from telemetry (flap_ratio changes vs flight phase)
ST-531	L3 Scoring: Stall Recovery	Grade stall recognition & recovery from stall_warn + pitch/airspeed data
ST-532	L4 Scoring: T&G + Emergency	Grade 5 T&G consistency + emergency landing on Rwy 06
ST-551	Walk-Around Scoring	Grade L3 walk-around from static item toggles + control sweeps + timing

12.2 MEDIUM Priority (Arc 2-4 Content)

Story	Title	Description
ST-524-526	Quiz Banks 251-253	Full question sets for Arc 2, 3, 4 check flights
ST-540	Arc 2 Content	L5-L7: objectives, briefs, scoring rubrics for pattern mastery
ST-541	Arc 3 Content	L8-L10: departure protocols, navigation, radio work
ST-542	Arc 4 Content	L11-L14: emergencies, weather, night flying
ST-550	Pattern Leg Classification	Detect crosswind/base/final by heading analysis

12.3 LOW Priority (Advanced Features)

Story	Title	Description
ST-543	Arc 5 Content	L15-L17: cross-country planning, dual/solo XC
ST-560	3D Flight Replay	Interactive 3D flight reconstruction viewer
ST-561	Real-Time Streaming	WebSocket telemetry for live flight monitoring
ST-570	Instructor Annotations	CFI can annotate specific telemetry moments
ST-571	Student Comparison	Compare two students' flights side-by-side
ST-572	Achievement System	Badges for milestones (first solo, night rating, etc.)

12.4 Content Specialist Interfaces

Future agents can populate content through these database fields:

Field	Model	Format	Used By
objectives	Lesson	JSON string[]	Brief tab
briefContent	Lesson	Text (markdown OK)	Brief tab
debriefTemplate	Lesson	Text	Debrief tab
scoringRubric	Lesson	JSON {focuses: string[]}	Scoring engine
questions	Quiz	JSON [{question, options[], correctIndex, explanation}]	Quiz tab
groundSchoolProductionNotes	Lesson	Text	Content tracking
flightRubricProductionNotes	Lesson	Text	Scoring dev tracking
mediaProductionNotes	Lesson	Text	Media production tracking
scoringTuningNotes	Lesson	Text	Scoring calibration tracking

Appendix A: Telemetry Field Reference

CSV Column	Type	Unit	Source	Notes
t_sec	float	seconds	Stock	Time since recording start
sim_time	float	seconds	Stock	X-Plane simulation time
lat, lon	float	degrees	Stock	WGS84 coordinates
alt_msl	float	feet	Stock	Altitude above mean sea level
alt_agl	float	feet	Stock	Altitude above ground level
ias_kts	float	knots	Stock	Indicated airspeed
gs_kts	float	knots	Stock	Ground speed
vs_fpm	float	ft/min	Stock	Vertical speed
pitch_deg	float	degrees	Stock	Pitch attitude (+ = nose up)
roll_deg	float	degrees	Stock	Roll angle (+ = right)
hdg_deg	float	degrees	Stock	Magnetic heading
aileron	float	-1 to 1	Stock	Aileron deflection
elevator	float	-1 to 1	Stock	Elevator deflection
rudder	float	-1 to 1	Stock	Rudder deflection
throttle	float	0 to 1	Stock*	*v4 fix: scalar not array
mixture	float	0 to 1	Stock*	*v4 fix: scalar not array
flap_ratio	float	0 to 1	Stock	Flap deployment (0=up, 1=full)
elv_trim	float	-1 to 1	Stock	Elevator trim position
rpm	float	RPM	REP	Engine RPM from REP
mp_inhg	float	inHg	Stock	Manifold pressure
on_ground	int	0/1	Stock	Weight on wheels
g_normal	float	G	Stock	Normal acceleration
slip_deg	float	degrees	Stock	Slip/skid ball deflection
stall_warn	int	0/1	REP	Stall warning active
chock_left	int	0/1	REP	Left wheel chock present
chock_right	int	0/1	REP	Right/nose wheel chock

tiedown_left	int	0/1	REP	Left wing tiedown
tiedown_right	int	0/1	REP	Right wing tiedown
tiedown_tail	int	0/1	REP	Tail tiedown
pitot_cover	int	0/1	REP	Pitot tube cover
head_yaw	float	degrees	Stock	Pilot look direction (left/right)
head_pitch	float	degrees	Stock	Pilot look direction (up/down)
view_type	int	enum	Stock	0=cockpit, 1026=REP walk-around
beacon_on	int	0/1	REP	Beacon light state
battery_on	int	0/1	REP	Battery master switch
avionics_on	int	0/1	REP	Avionics master switch

Missing columns default to 0. The parser never crashes on missing or unknown fields. Extra columns (e.g., rep_*) are captured as additional data.

Appendix B: Admin Operating Instructions

B.1 Login

1. Navigate to the app URL
2. Enter email: admin@s2s.com
3. Enter password: S2SA123
4. You'll be redirected to the Dashboard with an "Admin" button visible

B.2 Viewing All Lessons

As admin, ALL 17 lesson nodes are clickable regardless of gate status. Each lesson shows an  badge and content readiness flags.

B.3 Force-Completing a Lesson

1. Click "Admin" button on Dashboard
2. Go to "Students" tab
3. Find the student
4. Click "Force L#" button next to the desired lesson
5. Lesson is marked PASSED with score 5, quiz passed, debrief viewed

B.4 Toggling Content Readiness Flags

1. Admin Dashboard → "Lessons" tab
2. Each lesson shows 5 flags: Flight, Ground, Quiz, Scoring, Media
3. Click a flag to toggle it on/off
4. Lessons with `flightSchoolReady = false` appear as PLACEHOLDER nodes for students

B.5 Viewing Raw Telemetry

1. Navigate to any lesson → Debrief tab (requires at least one flight session)
2. Scroll to bottom — admin-only "Raw Telemetry Debug" panel is visible
3. Shows full JSON of parsed telemetry data and segment analysis

B.6 Uploading Telemetry for Testing

1. Navigate to any lesson → Flight tab
2. Click "Upload CSV" and select a `s2s_telemetry_*.csv` file

3. System will parse, analyze, and score the flight
4. Results appear immediately; debrief tab becomes available

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Screen to Sky Development Debrief v1.0

ST-511: Backbone Build Report

May 8, 2026

